

# CONTINUATION

## Continue — Project Status Snapshot

**Revision:** 27 **Last modified:** 2026-08-20T12:17:48Z

**TERMINAL STATE (this write) — 45 commits landed since Rev 26 (df7bc41 → e0d60ab): constitution v1.3.0 drift-review, BOB-126 fallout closed, quality-tooling 3-agent fan-out**

- **Staleness measured, not assumed (§11.4.6):** Rev 26 was written at df7bc41 (2026-08-19T17:25:34+02:00). HEAD is now e0d60ab (2026-08-20T12:01:57+02:00) — **45 commits, ~19h elapsed**, not the “53 days / 24+ commits” figure this session’s own workable item (BOB-081) was originally filed under; that figure described an earlier, since-superseded staleness state (Rev 25 or before). `git status --porcelain` on main at write time shows the tree otherwise clean except an in-flight parallel edit (see below).
- **3 remotes IN SYNC at HEAD e0d60ab** (github/origin/upstream — 0 ahead / 0 behind, verified via `git fetch --all + per-remote git log --oneline <remote>/main..HEAD` and the reverse).
- **In-flight, NOT part of this write:** `scripts/pre_build_verification.sh` (modified, uncommitted) + two new untracked files `scripts/lib/export_staleness.sh` and `tests/unit/test_export_staleness_oracle.sh`. These are consistent with the `TODO(EXPORT-SYNC-CONTENT-HASH)` migration plan e0d60ab left explicitly open (mtime-based export-staleness check is not reproducible across fresh clones — nanosecond mtime deltas + .html sorting before .md on checkout produced false-STALE on two fresh-clone runs). Owned by a concurrent session/agent per this task’s file-scope restriction; not committed, not evaluated here.

- **Constitution submodule bumped to v1.3.0** (78f18e8, pointer now 502586c) — full drift-review closing ~4 months of doc/code drift (container roster 2→5 services, plugin roster 12→43, cookies-file autoload, Hard Stop #3 orchestration ownership, Go webui-bridge profile, new Principles XII Anti-Bluff Captured Evidence + XIII Host-Session Safety) plus new universal anchor **§11.4.263 process-group signal-safety mandate** (the BOB-126 killpg(pgid≤1) forensic anchor, generalised across Python/Go/Rust/Bash/C).
- **BOB-126 fallout fully closed:** BOB-127 (evidence\_path fix), BOB-128 (killpg-carrier guard collision with the new §11.4.263 gate — fixed be36f32; CM-TEST-MOCK-PID-PATCHED-WHEN-REAL-PID landed as pre\_build invariant 29), BOB-129 (**real production slowapi/starlette rate-limiting defect** found by a Task-105 subagent and fixed on production endpoints, before/after evidence captured — 44f3bbe/e141d5b), BOB-130 (pre\_build timeout raised for a genuinely CPU-bound 93s fixture), BOB-131/132/133/135 registered. All eight registered/closed in the DB at d9955d6, exports regenerated.
- **5223f48 — 5 caveats closed with RED→GREEN evidence, not assertion:** (1) compute-badges.sh — a bare awk filter had been corrupting README by overwriting its own documenting prose line, plus a non-idempotent docs/TESTING.md regenerator that had accumulated to 7 stray blank lines; both fixed + guarded (test\_compute\_badges\_carrier\_match.sh). (2) A .gitignore rule was shadowing 58 TRACKED files (including the §11.4.109 hook config .claude/settings.json and production config/merge-service/\*), which made git add exit non-zero under set -euo pipefail — i.e. it could silently break the mandated commit entrypoint; unshadowed + guarded (test\_tracked\_files\_are\_addable.sh). (3) 2 dead constitution-mirror push URLs removed (repos verified non-existent via gh) — multi-URL fan-out was aborting on first failure and starving URLs 5-10; fan-out now completes 8/8. (4) QA discovery ledger: 2 new out-of-band entries with escape-audit + new-check, 18→20. (5) speckit.git.initialize root-cause fixed (an .specify/extensions.yml config omission). **Bonus finding:** tests/unit/\*.sh were being executed by nothing (ci.sh only collects test\_\*.py) — new pre\_build invariant 30 CM-BASH-UNIT-TESTS-EXECUTED wires them in; first run immediately surfaced 3 latent FAILs, quarantined by name as tracked debt per §11.4.226 (predicted exactly this).
- **e0d60ab (HEAD) — 3-agent fan-out, each verified by the controller, none accepted on claim (§11.4.262):** Agent 1 fixed a ~50%-flaky DOCX-export test (GNU vs BSD stat -f syntax divergence — portable portable\_mtime() helper landed, validated against real GNU/BSD polarity). Agent 2 stopped two challenge scripts from bumping mtimes of real tracked files on backup/restore (cp→cp -p), verified byte-identical + unchanged mtime/sha256 post-run. Agent 3 widened the mutation-residue “no-trace” assertion from 3 tracked paths to **3958** (main repo + constitution submodule — git ls-files does not descend into

submodules), replacing a whole-second stat -c%Y compare (proven blind to same-second mutate+restore) with a bash -nt-vs-marker-file compare, plus three new §11.4.201 refusal branches (empty corpus / unenumerable legacy path / unproven control needle) and a 6-quadrant polarity proof against the shipped bytes. Controller work: 8 stale .playwright-mcp tracked snapshots untracked (kept on disk, zero consumers verified, operator-approved per §11.4.122); docs/BOBA\_DATABASE.md's dangling § Key Rotation anchor fixed + the rotation procedure honestly marked NOT executable (no rotate-key/envfile-replace in source — corrects an earlier overclaim). QA ledger rev 15, entries 20→22, gate 32/32 passed. **One item deliberately left OPEN, not closed**, per §11.4.226 evidence-class discipline: EXPORT-SYNC-MTIME-NOT-REPRODUCIBLE-ON-FRESH-CLONE — see the in-flight files noted above.

- **Standing follow-ups from Rev 26 — status re-checked this write:** “helixqa nested dirty state” — **now CLEAN** (git status --porcelain inside submodules/helixqa returns empty at write time; superseded, no longer open). “Retrospective correction of BOB-116/120/123 incident docs” — **DONE** (b696873, landed same day as Rev 26, ~2 min after). “Constitution/gitverse push rejected by org policy” — **still open**, no new evidence found this write; remains a tracked §11.4.197 followup. “§11.4.185 manual QA before release tag” — **UNKNOWN**: no commit in the df7bc41..HEAD range references manual QA or §11.4.185; not confirmed done, not confirmed still pending — treat as still owed until evidence surfaces.
- **Open work-item tracker snapshot** (docs/workable\_items.db, queried live at write time): **50 non-terminal items** — 43 Queued, 3 In progress (BOB-066 Lava P3 upstream-proxy wiring, BOB-104 CodeGraph nested-.gitignore regression challenge, BOB-111 real rate limiting for boba's 3 public HTTP endpoints), 4 Operator-blocked (BOB-008 RuTracker CAPTCHA, BOB-077 second Auto-commit host identification, BOB-101 --profile go parity goal, BOB-102 LAN-exposure threat model — all await an explicit operator decision). The RD2-series backlog (BOB-065/068/069/070/074/076/078-100) covering egress diagnosis, the unattributed Auto-commit mechanism, DDoS/scaling/UX test-type coverage, and README doc-link completeness is still fully queued. Full enumerated list in docs/Issues.md / docs/Issues\_Summary.md (§11.4.57/§11.4.212).
- **This entry closes BOB-081** (RD2-14 — author this Session-N CONTINUATION.md entry). Test-count badges currently read 5358 Python / 371 frontend tests collected (README.md, refreshed at c37727b).

## TERMINAL STATE (previous write) — 7-forced-logout chain CLOSED + universal §11.4.263 anchor landed

- **Pre-build state:** full merge\_service unit sweep 863/863 PASS + Go 14/14 packages race-clean. Real captured evidence at docs/qa/2026-08-19-post-fix-verify/.
- **3 boba remotes IN SYNC at HEAD e984f4b:** 7-forced-logout chain resolution + DB update + regenerated Issues/Fixed/summaries.
- **BOB-126 7th forced-logout CLOSED — REAL ROOT CAUSE found + fixed:**
  - Kernel audit trail captured 2026-08-19 16:43:43:  
audit[399861]: SYSCALL syscall=62 a0=ffffffff a1=9  
comm="pytest" exe="/usr/bin/python3.14" = pytest calling kill(-1, SIGKILL).
  - tests/unit/merge\_service/  
test\_deadline\_tunable.py::test\_deadline\_hit\_flag\_true\_when\_readline\_times\_out created AsyncMock() without setting mock.pid as int. Production \_search\_public\_tracker called os.killpg(os.getpgid(proc.pid), SIGKILL). MagicMock.\_\_int\_\_ defaults to 1, so os.getpgid(1) == 1 → os.killpg(1, SIGKILL) → glibc → kill(-1, SIGKILL) = SIGKILL every UID-1000 process.
  - Bug existed since 2026-04-24 (~4 months). Prior 6 incidents (BOB-116/120/123/124/125) all misattributed to PAM/Linger contradiction — that was DOWNSTREAM effect.
- **Fix chain landed on all 3 boba remotes + 3/4 constitution upstreams:**
  - ad4b46a — boba: search.py int-guard on pid + pgid before killpg (both cleanup paths); test\_deadline\_tunable.py explicit mock.pid=12345 + patched killpg + new test\_bob126\_regression\_deadline\_path\_never\_calls\_killpg\_with\_pgid\_le\_1 §11.4.115 RED regression guard.
  - 502586c — constitution: NEW universal anchor **§11.4.263 process-group signal-safety mandate** covering Python/Go/Rust/Bash/C with four-layer coverage. Full text in Constitution.md, byte-identical compact mirrors in CLAUDE/AGENTS/QWEN/GEMINI.md (md5 542e30907eaa00ffb1949cd2fe528ba2). Pushed to gitflic + gitlab + github; gitverse rejected on org policy (deferred).
  - bf01cf3 — boba: constitution pointer bump.
  - 1b06858 — boba: BOB-126 incident doc Rev 4 + evidence bundle + BOB-095 refresh + wave-6 recovery.
  - d7da1af — boba: 5 BOB-122-fallout test fixes (4 iptorrents assertions + 1 SSE slowapi request-shape).
  - e389c29 — boba: QA\_DISCOVERY\_LEDGER Rev 12 (7-incident §11.4.238 escape entry).

- 0027dba — boba: BOB-124 Rev 2 retrospective correction (pointed at BOB-126 root cause).
- e984f4b — boba: workable\_items.db BOB-124/125/126 chain-close + regenerated Issues/Fixed + summaries + HTML/PDF + full evidence bundle.
- **Verification** (§11.4.5 captured evidence):
  - 863/863 merge\_service unit tests PASS in 103s (was 858/863 pre-fix)
  - 14/14 Go race-detector packages PASS clean in 20s
  - No 8th incident since fix landed (**50+ min elapsed vs prior ~37-38 min cadence** at 17:30)
  - Fixed.md now shows BOB-124 + BOB-125 + BOB-126 with proper resolution details
- **Standing follow-ups** (surfaced by subagent C's docs+DB audit):
  - **helixqa nested dirty state** — ? tools/opensource/anthropic-quickstarts, m tools/opensource/docling, ? tools/test-apps/rest-demo. Legit submodules in helixqa's .gitmodules, need git submodule update --init inside helixqa + operator decision on docling test-data drift. Out of scope for this session.
  - **Constitution/gitverse push** — org policy rejected push-to-create. Track as §11.4.197 followup.
  - **Retrospective correction** of BOB-116/120/123 incident docs (still carry pre-BOB-126 PAM/Linger hypothesis).
  - **§11.4.185 manual QA** — required before release tag. Operator decision on version bump.
- **§11.4.6 root cause CONFIRMED:** not PAM/Linger. Not systemd-oomd. Not idle timeout. A specific Python test called kill(-1, SIGKILL). Documented + fixed + universally guarded.

## Session commit summary — 2026-08-19 landings (18 commits, all pushed)

Rescue + forensics + watchdog + reviews + backlog: - 59edabd  
 BOB-118 badge computer partial-work rescue - 6c0b785 BOB-123  
 4th forced-logout PAM/Linger breakthrough doc - d635885  
 watchdog v1.0.0 (system.slice systemd unit + install/uninstall + challenge + README) - a9c2af9 Round-1 §11.4.209 fixes (I1-I4, M1-M7, NIT) - 69de436 Round-2 fixes (N1-N6) - 8278a42 Round-3  
 residuals — zero-finding GO reached - f517eaa BOB-118  
 completion: scripts/compute-badges.sh + unit test + doc - 0fc524a  
 QA evidence bundles (BOB-066/072/073/075/112/116,  
 task-78/80/85/e2e/rebuild/review + coverage-escape +  
 session-2026-08-16) - 652a4b4 docs/scripts/ + docs/testing/ export  
 twins - bedbcddc MERGE\_SEARCH regen + task-78 sig fixture  
 outputs + db-delta - <hash> badge refresh + 12 export twins (this  
 commit) - <hash> BOB-117 followup: remove residual “likely” from  
 plugin\_parse\_failure summary

## Sub-tasks CURRENTLY IN FLIGHT (3 subagents)

- Agent A → BOB-099: constitution/scripts/hooks/guard-forbidden-commands.sh substring-match false-positive fix + RED-first test
- Agent B → BOB-105: scripts/anchor\_block\_integrity\_check.sh + challenge + 4 fixtures + companion doc (§11.4.227(B) mechanical gate)
- Agent C → Task #87 anti-bluff correction doc + Task #42 §9.2 hardlink cross-device fallback

## Session commit summary (~20 boba commits + 4+ constitution commits since 457cca4)

## SESSION 2026-08-15 → 2026-08-18 (Session 4db6eadb-03e7)

**Working state at last write:** 3 boba remotes IN SYNC at 9c282ff. resource-pressure-signature-check systemd-user timer LIVE (next fire 22:42 CEST). 3 subagents in flight: Task #78 (IMPORTANT-1 real per-signature RED fixtures), Task #79 (IMPORTANT-2 SQLite differential-dump discipline), Task #80 (docs\_chain codegraph-status sync/verify contradiction).

## Major deliverables this session (in commit order)

- **Amendment round v68 landed** (Constitution + 4 mirrors + docs\_chain + boba pointer, 22 new anchors §11.4.239-262 + 5 extensions + §11.4.140/141 collision resolution → §11.4.255/256 re-mint)
- **Tag helixconstitution-v68** published on 4 platform releases (GitHub × 2 + GitLab × 2) + 8 constitution git remotes
- **BOB-064:** durable-run.sh helper (commit 20571a0)
- **BOB-066:** audit L1/L2/L4 wired (commit b9c2c50)
- **BOB-067:** Jackett cookie-login + HelixQA fake (commits 6231200 + 6dfc756)
- **Loader auto-discovery** + Jackett submodule bump v0.24.2353 → v0.24.2406
- **All 5 tracker credentials wired** (RUTRACKER, NNMCLUB, RUTOR, KINOZAL, IPTORRENTS) with §11.4.10.A pre-store leak audits
- **cookies\_.txt convention** at ~/Downloads with atomic auto-load into .env chmod 600

- **systemd integration** surviving reboot (boba.target + boba-stack.service + boba-webui-bridge.service, linger=yes)
- **Production install.sh** authored + LAN access verified on 192.168.1.90:{7185,7186,7187,7188,7189,9117}
- **BOB-112 boba-jackett /healthz DDoS mitigation:** sync.RWMutex TTL cache, live wrk **97.1% → 0.0% timeout** (400/412 → 0/812,149 requests) — commit e2a2e3e
- **BOB-113:** install-dev-tools.sh (wrk/hey/siege) — commit c7dfdde
- **Task #66 commit-push-all.sh -scope flag** (BOB-068 sweep-pattern interim fix) — commit 0972cbc
- **BOB-108:** workable-items export Revision-counter regression fix — constitution 4a17867 + boba 855ce65 + 3520621
- **CodeGraph 1.5.0 nested-.gitignore regression:** upstream issue #1567 + draft PR #1568 at colbymchenry/codegraph
- **Task #72 docs\_chain unpushable-gitlink:** cherry-pick + push resolution (docs\_chain 8ccf505 + constitution 774ac57 + boba a4c6c7a + 41da88a)

## **CRITICAL 2nd forced-logout incident (2026-08-18 20:50:59)**

- CONST-033 triage clean: no kernel OOM, no systemd-oomd trigger, no lid-suspend, no forbidden mechanism
- §12.12 EAGAIN cascade signature at 20:45:48 confirmed (jackett SocketException (11) to iptorrents+kinozal+rutracker simultaneously)
- 15 GB pathological ugrep reaped post-relogin: ugrep -o with . \{0,120\} variable-length context + 3-way alternation on 14K-line CLAUDE.md — freed 16 GB instantly, PSI Avg10 1.77 → 0.08
- §11.4.6 UNCONFIRMED: exact SIGKILL source unattributable from systemd journal — PENDING FORENSICS (task #79)
- **Fix commit 1f42357:** CONST-033 challenge false-positive fix + NEW resource\_pressure\_signature\_challenge.sh (5 signatures, §11.4.115 polarity) + docs/incidents/2026-08-18-perceived-forced-logout-2nd.md + 8 evidence artifacts
- **§11.4.238 ledger entry 98412bf** — FORCED-LOGOUT-2026-08-18-2ND (Rev 6→7, count 14→15)
- **Task #77 wire commit ecb3bfe:** pre\_build invariant 25 CM-RESOURCE-PRESSURE-SIGNATURE-CHECK + hourly systemd-user timer INSTALLED + ARMED (next fire 22:42 CEST)
- **Persistent memory playbook** at ~/.claude-claude4/.../memory/forced\_logout\_incidents.md
- **Task #41 BOB-068 investigation a7e55f9:** verdict = NO DAEMON. Root cause = shared-checkout race on default git add -A + shared docs/workable\_items.db. §11.4.179 clone-isolation (Task #67 proposal) confirmed correct architectural fix.

## §11.4.209 review verdict: NO-GO with 3 IMPORTANT + 6 MINOR + 2 NIT

- **IMPORTANT-1** (BOB-116 polarity claim threshold-only, not real-signature — initially referenced as BOB-076 informal label, corrected 2026-08-18; BOB-076 is a distinct unrelated Type=Task DB item, see docs/incidents/2026-08-18-perceived-forced-logout-2nd.md): fix subagent in flight (task #78)
- **IMPORTANT-2** (3520621 DB commit missing differential SQLite dump): fix subagent in flight (task #79)
- **IMPORTANT-3** (HTTPS git URL in install-dev-tools.sh): FIXED inline commit 9c282ff (both real invocation + advisory message → git@)
- MINOR findings tracked as followups
- §11.4.209 Fable-xhigh re-review OWED (this review ran on opus fallback per Agent-tool effort=? §11.4.231(F)(b) honest boundary)

### To resume

- 3 subagents in flight: notifications will arrive
- New commits will push automatically (their briefs specify commit + push)
- Task tracker #40, #42 remain deferred (need operator design decisions)
- Constitution submodule pointer: constitution at 4a17867 (BOB-108 fix) → boba 9c282ff → docs\_chain 8ccf505
- Everything else in Session 15 CONTINUATION section below remains valid history **Session:** 2026-08-10/11 (Session 16 — HUGE session: pre\_build brought to 25/25 GREEN with six new invariants (INV0/19/20/21/22/23), §11.4.238 QA-discovery gates wired, §11.4.84 subagent-mutation fence Layers 1+3 landed, workable-items diff false-positive fixed + new correct-history- evidence subcommand, 39 BOB backfills closing a §11.4.197 mass violation, three stress+chaos suites landed GREEN, HelixQA banks parametrized for RD2-34, QA\_DISCOVERY\_LEDGER at Rev 3 with a new automated-background-scan channel. Definitive host-kill root cause identified — ALT Linux KillUserProcesses=true compile-time default; mitigation config staged at scratchpad/20-kill-exclude-milosvasic.conf but STILL BLOCKED on operator sudo.) **Last commit (boba main):** 1d0268f (chore(hygiene): gitignore scratchpad/ + update helixqa submodule URL to renamed upstream). Prior big session-16 commit: 4b7b21d (session-16: pre\_build 25/25 GREEN — §11.4.238 gates + §11.4.84 fence + backfills + stress+chaos + BOB-009/010 fix). **Branch:** main **Working tree:** CLEAN (post-session-16 commits landed + pushed). Only in-flight edit: this CONTINUATION.md revision itself.



Send continue to pick up exactly where we left off. This file is the single source of truth for session handoff (§12.10 / §11.4.131).

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## **CURRENT STATE — Session 16 (2026-08-10/11): pre\_build 25/25 GREEN + §11.4.238/§11.4.84 hardening + 39 BOB backfills + stress+chaos + BOB-009/010 fix**

**Operator asked for the §11.4.238 QA-discovery regime to actually bind (per its zero-grace-period “no escape hatch” clause) and for the §11.4.197 mass-violation from Session 15’s audit backlog to be closed.** Delivered as one big commit (4b7b21d, 37 files) + a hygiene follow-up (1d0268f), both pushed to all upstreams on boba main. Constitution + HelixQA submodule pointers advanced + pushed. This session survived multiple kill events (below) — the constitution submodule and HelixQA submodule work landed BEFORE the boba-side work, so no in-flight cross-repo saga was left half-applied (§11.4.191 preserved).

### **What landed (Session 16 — all published)**

**boba main-repo (4b7b21d + 1d0268f): - Pre-build 25/25 GREEN** — six NEW invariants in scripts/pre\_build\_verification.sh + scripts/pre\_code\_review.sh: **INV0** CM-PREFLIGHT-INTERPRETER (catches bare python3 -m pytest vs .venv/bin/python ABI drift at collection time — RD2-41b/RD2-43); **INV19-21** §11.4.238 (ledger fresh + counts aligned, every out-of-band entry carries escape-audit: + new-check:, propagation literal in the §11.4.157 mirror set — INV21 wrong-seam bug corrected to check constitution/{CLAUDE,AGENTS,QWEN,GEMINI}.md, not project root); **INV22** CM-DOCS-CHAIN-STEP1-REAL-INVOCATION (§11.4.238 RD2-41a retroactive catcher, anchored ^\s\*ERROR: grep, structure-not-substring per §11.4.201(7)(a)); **INV23** CM-NO-PRODUCTION-MUTATION-RESIDUE (§11.4.84 layer 3, string-concat de-mutation so scanner doesn’t self-match, FIXTURE\_ROOT env override for golden-good/golden-bad). pre\_code\_review.sh: mutation-marker carrier false-positive fixed 36→0 hits. - **§11.4.84 subagent-mutation fence** (crypto.go RED-window incident from Session 15/16): **Layer 1** — subagent briefing template with 3 mandatory isolation patterns (git worktree / atomic mutate-run-restore / mutation tool); **Layer 3** — INV23 above. **Layer 2** (post-tool-use hook) DEFERRED pending runtime capability check; design at scratchpad/task-20-84-fence-design.md. - **§11.4.238 QA discovery ledger** (docs/QA\_DISCOVERY\_LEDGER.md Rev 3): NEW **INC-2026-08-10** entry — crypto.go GCM auth-bypass mutation left in production file during

Agent H's RED window, caught by background security-review plugin (a new discovery channel: `automated_background_scan`). Full escape-audit + new-check documented. Table split 7 → 8 out-of-band. - **workable\_items.db reconciled + expanded** (63 → 102 items): **39 BOB backfills** for ~40 RD2-/GA-/RW-\* audit-doc items that had no tracker rows (Session 15's §11.4.197 mass violation — CLOSED). BOB-064..067 for Lava P1..P4 (RD2-15 reservation); BOB-068..102 for remaining audit items. **BOB-009 + BOB-010** evidence\_path corrections via a NEW workable-items correct-history-evidence subcommand (constitution submodule) — no more raw SQL, no more falsely-implied reopen. workable-items diff false-positive fixed (bare --db invocations were flagging every row as absent-from-MD). validate: OK — 102 items. - **Stress+chaos tests (§11.4.85)**: NEW tests/stress/test\_tracker\_fetch\_stress\_chaos.py (9/9 GREEN, 27.5s, 9 JSON evidence files under qa-results/stress\_chaos/); NEW qBitTorrent-go/tests/integration/scheduler\_hooks\_sse\_stress\_chaos\_test.go (6/6 GREEN + -race clean, 5.6s); +702 lines in qBitTorrent-go/tests/integration/jackett\_db\_test.go — 5 new chaos cases (byte-corruption, concurrent-writer, WAL-sidecar, master-key-rotation, mid-tx SIGKILL), 5.8s -race clean. - **Hygiene follow-up** (1d0268f): .gitignore scratchpad/ (24 files/176K session artifacts, §11.4.10 scan clean); .gitmodules helixqa URL updated HelixQA.git → qa.git following upstream repo rename (GitHub warned "This repository moved"), git submodule sync propagated.

**constitution submodule** (pushed to all 8 remotes): 4a2adac (fix workable-items diff false-positive + new correct-history-evidence subcommand + sudo/su quote-aware guard) + 03f0b0c (merge remote 6f960ca design-toolkit pointer bump per §11.4.188).

**HelixQA submodule** (4289cf4, pushed): banks(boba) — added BOBA-PRX-009/010 RD2-22 Challenges + parametrized /Volumes/T7 hardcoded paths (RD2-34 closed).

## **Session-kill root cause — DEFINITELY IDENTIFIED, mitigation staged, BLOCKED on sudo**

- **Root cause** (per Session 15's live busctl finding): ALT Linux ships systemd-logind with the **compile-time default KillUserProcesses=true** — every GDM/GNOME session close SIGKILLS the entire user@1000.service slice (tmux, Claude Code, everything) regardless of Linger=yes. Confirmed multiple hits again this session.
- **Mitigation staged**: scratchpad/20-kill-exclude-milosvasic.conf (a logind.conf.d drop-in for KillExcludeUsers=ilosvasic — narrower than blanket KillUserProcesses=no). §11.4.6 HONEST GAP: the file was described in the session hand-off brief but is NOT visible in the scratchpad listing at write time — either it was staged elsewhere, wiped by a kill event, or the brief refers

to a design not yet materialized on disk; next session MUST re-verify + author the drop-in fresh if absent.

- **Blocked on:** sudo cp of the drop-in into /etc/systemd/logind.conf.d/ + systemctl restart systemd-logind. Per repo CLAUDE.md § “Host Power Management — Hard Ban” AND the constitution’s §11.4.109 PreToolUse guard, the agent’s sudo is CORRECTLY refused; the operator must run the copy + restart themselves (the !-prefix-in-permissions path is the sanctioned operator-executed channel). Until then, every session remains at risk of the same abrupt kill.

## Pending operator §11.4.66 decisions

1. **Apply the KillUserProcesses fix** (narrower KillExcludeUsers=ilosvasic vs blanket KillUserProcesses=no) — see above. HIGHEST priority: every future session is at risk until this lands.
2. **§11.4.84 Layer 2 post-tool-use hook** — runtime capability check owed; design at scratchpad/task-20-84-fence-design.md.
3. **RD2-22 live-verify still owed** (source fix done Session 15; the full tests/contract/+tests/unit/ regression sweep was killed mid-run by the logind issue and still needs to complete + a live curl-verify 401/200 on a running container + the §11.4.135 regression guard + HelixQA Challenge entry).
4. **§11.4.234 dedicated commit/push wrapper** (task #25) — deferred with operator go-ahead; direct commits used this session.

## In-flight work carried forward

- **pytest sweep v5** — the RD2-22 regression sweep still owed (pending item 3 above).
- **Task #27 (suite-scope repro)** — still needs a scope-widened reproduction pass. Details in the operator brief that opened this session; re-derive from docs/workable\_items.db + Session-15 audit if the working note is not in scratchpad.

## What to do FIRST on next-session pickup

1. **Land the logind fix** (mitigation config → /etc/systemd/logind.conf.d/ + systemctl restart systemd-logind). Verify with busctl get-property org.freedesktop.login1 /org/freedesktop/login1 org.freedesktop.login1.Manager KillUserProcesses (should now show false OR the user should now appear in KillExcludeUsers). Every subsequent step depends on the session no longer being at risk of a mid-work kill.
2. **Complete RD2-22 live-verify** — resume the killed regression sweep, curl-verify 401/200, land the §11.4.135 guard + HelixQA Challenge entry.

3. **Resume docs/GOVERNANCE\_AUDIT\_2026-08-08\_ROUND2.md** remaining root-cause-grouped items in priority order; the RD2-40 §11.4.238 compliance item is now **MECHANICALLY ENFORCED** as of this session (INV19-22), but the ledger's ongoing curation is the standing work.
- 

## **CURRENT STATE — Session 15 (2026-08-08): governance re-audit + P0 security fix + host root-cause + multi- track kickoff**

**Operator asked for a full deep-research constitutional-compliance audit** (rules followed, constitution-derived technology incorporated, gaps/danger-zones/inconsistencies identified, systematic root-cause remediation plan with live verification). Delivered as docs/GOVERNANCE\_AUDIT\_2026-08-08\_ROUND2.md (Rev 3) — live re-verification of every item in the 2026-08-07 audit (GA-01..27) + the original 2026-06-14 plan (RW-01..21), a fresh gap scan (RD2-01 through RD2-09), and root-cause forensics on two previously-unattributed “Auto-commit” commits. **Read that document for the full, prioritized, root-cause-grouped remediation plan** — this entry only summarizes what changed THIS session.

### **What landed (Session 15)**

- **Governance audit, Round 2 (docs/GOVERNANCE\_AUDIT\_2026-08-08\_ROUND2.md)**. Confirmed **DONE** since the 2026-08-07 audit: Python/frontend/extension toolchains (4340 unit tests now collect cleanly), ruff down to 0 real violations (submodule-scoping fix), HelixQA bank symlinks fixed, start.sh --reload-python/--reload-plugins/--recreate real working implementations, CONST-033 triage doc, jackett-autoconfig dead-test removal. Confirmed **STILL OPEN**: 3 test files mislabeled integration/e2e/contract that mock the system under test, 14 #legacy-untriated tags, Go backend parity (operator-decision, unchanged), stress/chaos coverage gaps, HelixQA hardcoded /Volumes/T7 paths, docker-compose.quality.yml missing container-hygiene limits, a .trivyignore.yml with placeholder CVE IDs, DDoS-class testing fully absent. **NEW P0 finding (RD2-40)**: boba does not yet comply with the just-landed constitution anchor §11.4.238 (no manual-discovery ledger, no coverage-escape-audit process) — every finding in this very audit round is itself an instance of the out-of-band-discovery channel that anchor targets.

- **RD2-22 (P0 security) — source fix DONE, live-verify still owed.** PATCH /api/v1/schedules/{id} and PUT /api/v1/theme were unauthenticated even with BOBA\_API\_TOKEN set. TDD RED (tests/security/test\_hooks\_schedules\_auth.py extended) → GREEN (Depends(require\_api\_token) added to both routes in download-proxy/src/api/scheduler.py + routes.py) → 31/31 security suite + 176/176 theme/scheduler regression tests pass. A root-cause function-ordering bug (require\_api\_token referenced before its own definition, crashing the whole API module) was found and fixed during GREEN — require\_api\_token moved earlier in routes.py. **Still owed:** the full tests/contract/+tests/unit/regression sweep (interrupted mid-run by a session kill — see below) and a live curl-verify (401 unauth / 200 with-token) on a running container.
- **Host root-cause found: KillUserProcesses=true.** The operator reported the session/tmux/ Claude Code process being repeatedly killed. Systematic-debugging (journalctl OOM-killer check → zero hits; systemd-oomd → no kill-decision log lines; live busctl read of org.freedesktop.login1.Manager.KillUserProcesses → confirmed true) found the real mechanism: every time the local GDM/GNOME session closes, systemd-logind SIGKILLS the ENTIRE user@1000.service slice — tmux server, Claude Code, everything — regardless of Linger=yes. Confirmed 4 occurrences in one day via journalctl, one explicitly listing pid ... (claude) in its kill list. **Awaiting operator decision:** a logind.conf.d drop-in (KillUserProcesses=no blanket, or KillExcludeUsers=milosvasic narrower) — not yet applied, host-wide change, needs explicit go-ahead.
- **Track 11 — multi-track identity permanently adopted.** docs/MULTITRACK.md + scripts/multitrack/track\_branch\_label.sh (boba-owned wrapper around the constitution's shared, inherited-by-reference labeler — never edited in place). This checkout is registered as the home of Track 11; trunk (main) work stays T1 per the constitution's own hard-coded TRUNK RULE (never overridden); non-trunk work from this checkout labels T11. Execution stays single-track for now per operator instruction — no parallel ruler/ subagent streams spawned.
- **New constitution anchor §11.4.238** (“automated QA must be the DISCOVERER, not the confirmer” — manual QA must find zero new issues). Landed via a **live cross-session collision:** a parallel Claude session (Opus 5, same host observed via a +0500 timezone offset) independently authored and published the same anchor number first; this session's own unpublished duplicate was discarded (never reached any remote, safe to abandon) per §11.4.227(B)'s anchor-collision rule. constitution submodule pointer bumped + pushed (50c7a66). This ALSO resolved the mystery “Auto-commit” pattern from the 2026-08-07 audit — it is very likely this same second session/device, not an unknown external actor.
- **workable\_items.db reconciled.** BOB-008's DB↔MD body drift fixed (sync md-to-db) plus a genuine §11.4.148(D3) finding surfaced and fixed along the way (the Operator-Block-Details

UNBLOCK clause needed enumerated [A]/[B] choices, not free prose). BOB-009/BOB-010's `evidence_path` violations (narrative text instead of a resolvable path) were investigated — real evidence identified via git archaeology (commit 0558399; `scripts/boba-ctl.sh` + `scripts/docs_chain.sh` (renamed 2026-08-15 BOB-104 to `scripts/workable-items-export.sh`) still exist and back the claims) — but the `workable-items` tool has **no subcommand to correct a historical `item_history.evidence_path`** on an already-Completed item; raw SQL was correctly avoided (violates the tool-only-mutation discipline) and a reopen-then-close cycle was correctly avoided (would falsely imply the underlying work was broken). Tracked as an honest, open tooling gap, not silently worked around.

## In flight / NEXT ITEM (P0, do this first)

1. **Finish RD2-22:** re-run the full tests/contract/+tests/unit/regression sweep to completion (it was killed mid-run by the `KillUserProcesses` issue), then live curl-verify 401/200 on a running container, then add the \$11.4.135 regression guard + HelixQA Challenge entry.
  2. **Operator decision needed:** apply the `KillUserProcesses` fix (blanket vs. narrower — see above) — every future session remains at risk of the same abrupt kill until this lands.
  3. Work through docs/GOVERNANCE\_AUDIT\_2026-08-08\_ROUND2.md's remaining root-cause-grouped items in priority order (Root Causes 3-6 + the ungrouped P1/P2 items), starting with RD2-40 (\$11.4.238 compliance) given its zero-grace-period “no escape hatch” clause.
  4. Deferred until the whole backlog above is genuinely clear (operator's own gating condition): full production release — install script, user-level systemd/systemctl integration so the stack boots with the system, full docs, then notify the operator it's ready to use.
- 

## CURRENT STATE — Session 14 (2026-06-16): search fixed end-to-end + proven on nezha; one P0 in flight

Operator-reported “search is broken / crashing a lot” + “`rutracker/nmclub` never authenticate” — **ROOT-CAUSED and FIXED, proven live on the distributed-boot production stack (`nezha.local`, `podman`)**. Definitive end-to-end proof: full-fleet the matrix on nezha returned **2600 results, 23/29 trackers contributing, encoding-crashed: NONE**, all four private

**trackers authenticated** (rutracker 50 / nnmclub 50 / kinozal 50 / iptorrents 49). QA evidence: docs/qa/search-fix-verify-20260616/ (results.md + html/pdf/docx). BUGFIXES.md Rev 19 entries 48-55.

## **What landed (Session 14 — all published at HEAD 646b295)**

- **Multi-word query URL-encoding crash, 17 nova3 plugins** (dbd3858 7 maintained + da7d709 10 adopted) — a literal space crashed urllib (URL can't contain control characters); single-word worked, multi-word silently dropped ~17 plugins → “search broken / crashing”. Fix: per-plugin percent-encoding. **Verified live** (nezha): scoped the matrix 592, full-fleet 2600, NONE crashed. (BUGFIXES #48)
- **plugin\_bad\_query\_encoding honesty classifier** (33d90f2) — stopped reporting a self-inflicted encoding bug as a remote plugin\_crashed (§11.4.6). (BUGFIXES #49)
- **Private-tracker cookie auth** — RUTRACKER\_COOKIES injection bypasses the CAPTCHA-walled login (2fc29fc) + existing NNMCLUB\_COOKIES; cookies via scripts/extract-tracker-cookies.sh (§11.4.10). **Verified 50/50** on nezha. (BUGFIXES #50)
- **/auth/status cookie reflection** (9c2f8dc) — chips now show cookie-authenticated trackers green before the first search. (BUGFIXES #51)
- **JSON /api/v1/healthz** (137d7ff) — was swallowed by the SPA catch-all (returned HTML); now JSON. (BUGFIXES #52)
- **scripts/deploy-remote.sh — 4 silent-failure bugs** (d5b58cb / 9e059d3 / 42cdb02 / e6b9f8f) that kept fixes from landing on the remote (§11.4.108): inline-YAML-comment path, rsync abort on container-owned config/, py\_compile false “Syntax: Invalid”, and install-plugin targeting only qbittorrent not qbittorrent-proxy. Pipeline now completes [1/5]→[5/5] clean; re-verified via the pipeline (not manual cp). (BUGFIXES #53)
- **kickass §11.4.112 Won't-fix** (f3e7a4f) — 403 behind a Cloudflare/JS challenge, structurally unsolvable by a headless nova3 plugin; documented in docs/research/. (BUGFIXES #54)
- **Regression baseline: full unit suite at HEAD 4277 passed, 0 failed** (up from 4216 pre-session, zero regression).

## **Host note**

- **Host disk-full** observed this session (T7 / /Volumes/T7/tmp is the temp dir for this work). Keep an eye on free space before any container rebuild or large deploy — a full disk silently fails rsync/builds.

## **In flight — NEXT ITEM (P0, do this first)**

- **UTF-8 / Cyrillic query crash in 15 plugins (OPEN — NOT yet fixed, §11.4.6)**. A sibling defect class to #48 surfaced

during the #48 discovery extend-pass (§11.4.146/§11.4.118): the percent-encoding fix addressed the literal-space control-character crash; the non-ASCII (Cyrillic) byte path is a DISTINCT crash. RED reproduction test is in the working tree (untracked): tests/unit/test\_plugin\_unicode\_query\_encoding.py. **NEXT:** complete per-plugin UTF-8-safe encoding across the 15 affected engines, flip RED→GREEN (§11.4.115), run the §11.4.146 extend-pass to enumerate the full affected set, verify live on nezha, then complete BUGFIXES #55 with the GREEN proof + commit hash.

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## **CURRENT STATE — Session 13 (2026-06-13): stack booted + qBittorrent 5.x compat**

**The #1 release blocker is CLEARED.** The full multi-container stack is BOOTED and the live detect→send→torrent round-trip is GREEN — the synthetic infohash is independently confirmed in qBittorrent's real torrent list, then cleaned up (§11.4.14).

- **Boot blocker (macOS/podman) — root-caused + fixed.** / Volumes/T7 (external SSD) was not shared into the podman applehv VM → podman create hung on an unreachable bind path (proven: py3.13 + 3.14 hung identically; NOT a python-version issue). Fix (operator-authorized §11.4.122): recreated the machine with --volume /Volumes/T7:/Volumes/T7 --cpus 4 --memory 6144; export QBITTORRENT\_DATA\_DIR=/Volumes/T7/Projects/Boba/downloads; ./start.sh. All 5 containers healthy. Earlier boba-ctl up -d boot bug fixed in 8b6d245.
- **3 REAL product defects (qBittorrent v5.2.1) — 0dc247b.** The proxy spoke legacy text (200 "Ok.") while linuxserver:latest speaks 5.x JSON/204. download-proxy/src/api/routes.py: \_qbit\_login\_succeeded (204+QBT\_SID cookie, all 4 login sites) + \_qbit\_add\_succeeded (modern JSON add + 409-duplicate idempotency, all 3 add sites). Guard: tests/unit/test\_qbit\_login\_compat.py (19 tests). Independent code-review (§11.4.142/§11.4.134): 2 BLOCKING + 1 warning caught → clean GO. Documented BUGFIXES.md Rev 14 entries 36-38.
- **macOS LAN access** — ef121e7: scripts/ensure-macos-tunnel.sh now supports TUNNEL\_BIND\_ADDR=0.0.0.0 for LAN-IP access (default 127.0.0.1 unchanged).
- **Manual testing (stack UP):** http://localhost:7187/ (dashboard) · http://localhost:7186 (qBittorrent admin/admin) · :7189 boba-jackett · :9117 Jackett. Re-tunnel after reboot: bash scripts/ensure-macos-tunnel.sh (TUNNEL\_BIND\_ADDR=0.0.0.0 for LAN).
- **Remaining (operator-gated):** headful MV3-load e2e (needs a real display); store assets/submission.



- **Process note:** a background integration suite (test\_live\_containers.py, which does compose down/up) tore down the live stack mid-session; killed + restored. Parallel work during operator testing MUST stay hermetic (§11.4.119) — no integration/e2e/container-touching tests.
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## CURRENT STATE — Session 10 (BobaLink browser extension)

**New major feature in flight: the BobaLink browser extension. Plan + discovery + Phase 1 foundation + backend BE-1/BE-2 + a token-auth security fix landed. The Python unit suite stays FULLY GREEN at 4149 passed.**

### What is BobaLink

A WXT + TypeScript Manifest-V3 browser extension that detects magnet links and .torrent URLs on any page and forwards them to the running Boba merge service on port **7187**. Backend contract: POST http://<host>:7187/api/v1/download. The full plan, discovery analysis, and traceability live under docs/browser\_extension/ — master plan IMPLEMENTATION\_PLAN.md (9 phases), analysis artifacts \_analysis/01–06, planning artifacts \_plan/A–F, and the 245-item traceability matrix \_plan/C-traceability-matrix.md (240 v1 / 5 v2, proving nothing is skipped per §11.4.118).

### What landed (Session 10 — BobaLink)

- **b2356ae** — BobaLink master implementation plan + discovery/planning artifacts (docs/browser\_extension/); also bumped HelixQA submodule bcac236 → 4d2dcb2.
- **33a9815** — extension **Phase 1 foundation** scaffolded (extension/ — WXT config, TS, shared libs/types under extension/src/); **71 anti-bluff Vitest unit tests** (vitest 71 passed, eslint clean — crypto imports the REAL module, not an inline copy).
- **d46bffb** — backend **BE-1** (CORS for extension origins) + **BE-2** (raw .torrent upload endpoint) added to the Python :7187 API.
- **284d1c4** — bumped HelixQA submodule 4d2dcb2 → bca3b36; new Challenge bank submodules/helixqa/banks/boba-bobalink.yaml (6 cases).
- **192b945** — **security fix:** env-gated shared-secret BOBA\_API\_TOKEN enforced on the three :7187 download-write endpoints (when the env var is set, requests must carry a matching X-Boba-Token / Bearer token; behaviour unchanged when unset).

Verified by **15 token-auth tests** (tests/unit/api\_layer/test\_download\_token\_auth.py).

## What landed since Session 10 (BobaLink — Phases 2 & 3 + Phase 4 leaf)

- **7225470** — Phase 2 wave-1: parsers (bencode.ts/magnet.ts) + scanner base/site-db.
- **fa03323** — Phase 2 wave-2: .torrent-file SHA-1 infohash + link/text scanners + perf/stress.
- **946c61e** — Phase 2 complete: scanner orchestrator (cross-scanner dedup).
- **2e59572** — lint test files clean; lint script extended to tests/.
- **e8fde43** — Phase 3 shell + Phase 4 api leaf: api/{boba-client,queue} + content/popup/options.
- **15a9a61** — Phase 3 capstone: background service worker (message router). **HEAD**.

## Suite status

- Full Python unit suite: **4149 passed** — pytest tests/unit/ --import-mode=importlib (Session 10 baseline; unchanged this session). Evidence: qa-results/tokenauth\_fullsuite\_\*.log.
- Extension Vitest corpus: **379 passed across 33 spec files** — same-session npx vitest run (Tests 379 passed (379)); tsc --noEmit clean; npm run lint 0/0. (+92 over the 287/22 baseline.) Build: npx wxt build → **loadable .output/chrome-mv3/** (§11.4.38 — 8/8 manifest assets verified present).

## Current phase + next steps (Session 11 — two parallel waves landed; reviewed checkpoint)

- **Phases 1, 2, 3 + extension shell + background SW: COMPLETE.**
- **WXT build wiring: COMPLETE** — entrypoints at src/entrypoints/{background.ts,content.ts,popup/index.html,options/index.html} (thin wrappers); npx wxt build → loadable .output/chrome-mv3/; matches derived from SITE\_SELECTORS (24 hosts, no <all\_urls>); least-privilege manifest; §11.4.38 verified.
- **Phase 4 (backend integration): IN PROGRESS** — api leaf @e8fde43; **Phase-7 decrypt-before-send WIRED** — BobaClient.create() decrypts the encryptedBobaApiToken bundle; background reads the session passphrase from chrome.storage.session, sends decrypted plaintext, default-open when locked (RED→GREEN; token/passphrase never logged). **PENDING:** live-7187 integration (require\_backend(7187)) + detect→send→torrent-in-qBittorrent E2E on the real backend.
- **Phase 5 (tab groups): IN PROGRESS** — standalone src/tabgroups/index.ts (dedupe-across-group + batch dispatch) +

13 tests landed; **PENDING**: wire into background  
MENU\_SEND\_GROUP + add tabGroups/tabs perms (least-privilege  
review first).

- **Phase 6 (i18n): IN PROGRESS** — locale.test.ts guards en-catalog completeness. **Phase 7 (security): IN PROGRESS** — decrypt path + tests/security/\* (least-privilege/CSP/no-hardcoded-secret/secret-storage). **Phase 8: IN PROGRESS** — 379/33 green; real-module Challenge challenges/extension/detect\_and\_forward\_challenge.sh (mutation-verified); HelixQA BOBA-LINK-007; e2e is an honest operator-gated SKIP (sandbox can't load extensions). **Phase 9: PENDING** — ci-ext.sh gate + per-store packaging + §11.4.65 doc siblings.
  - **Next actions:** (1) **independent-subagent code-review re-pass** for the correctness/security/build lenses once the platform subagent-dispatch throttle clears (done in-context this session; re-run before any release tag per §11.4.40); (2) Phase 5 integration into background + perms; (3) Phase 4 live-7187 integration + E2E; (4) Phase 9 packaging; (5) regenerate Status\_Summary.md + §11.4.65 HTML/PDF siblings for the browser\_extension Status docs. Status: docs/browser\_extension/Status.md (Rev 2, accurate). QA evidence: docs/qa/bobalink-2026-06-10-session11/.
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## CURRENT STATE — Session 10 (2026-06-10, subagent-driven loop)

**Suite stays FULLY GREEN. This session was hardening + verification, subagent-driven (§11.4.70), every change code-reviewed before commit (§11.4.125/§11.4.134).**

### What landed (Session 10)

- **test\_credential\_env\_wiring.py isolation fix** — resolved the Session-9 “known low-priority follow-up” below. A merge\_service fake-namespace stub was installed at module-collection time with no teardown, so conftest's \_isolate\_download\_proxy\_modules captured-and-restored the *fake* stub into siblings in some run orders. Fixed by moving stub install into a search\_mod fixture that snapshots/restores the 3 injected keys (merge\_service, .search, .retry) in finally.  
**Evidence:** RED reproduction (the 5 credential tests + a temporary leak-probe) went 1 failed → all green; the file's own **5 tests pass** standalone; full unit suite **4121 passed** × seeds default/42/31337 (Session-10 run). Code-review **GO** (proved negation; no weakened assertions; ruff clean).
- **docs/qa/BOB-008/operator\_runbook.md** — copy-pasteable operator unblock procedure for BOB-008 (cookie path preferred; CAPTCHA path documented with its cap\_sid/cap\_code\_field

friction), every endpoint claim cited to `auth.py:line` (§11.4.83/§11.4.99).

- **Latent-leak discovery sweep (§11.4.118)** — audited all of tests/ for the module-level fake-stub-no-teardown pattern. Finding: the pattern is a **structural no-op wherever the test dir has an `__init__.py`** (real package pre-loads → `setdefault` never installs the fake), so the ~26 tests/unit/merge\_service/matches and `test_tracker_stats.py` are **inert** (no RED reproduces; strict TDD → no fix). The single genuinely-uncovered root is `pirateiro` (see follow-up).
- **Constitution inheritance verified (§11.4.32/§11.4.35)** — PASS: pointer present in both layers, propagated anchors present in canonical source, no conflict markers, `pre_build_verification.sh` 18/18 green.
- **pirateiro test-isolation fix — BOB-063 (operator-approved, implemented)** — tests/unit/test\_plugin\_pirateiro.py injected `sys.modules['pirateiro']` at module scope with no teardown; `pirateiro` was the one root uncovered by `conftest`'s isolation, so it leaked into later tests. The naive “add to `_POLLUTING_ROOTS`” does NOT work (the leak is already inside each per-test snapshot); the real fix caches+re-registers+purges the stub per unit test. **Evidence (§11.4.115):** RED 1 failed, 44 passed → GREEN 45 passed; negation proof (disable the purge → re-fails); full suite **4122 passed × seeds default/42**. Standing regression guard tests/unit/test\_pirateiro\_isolation\_guard.py (§11.4.135). QA at docs/qa/BOB-063/evidence.md. **Closed BOB-063** (Task → Completed).
- **Constitution submodule advanced 60e2d66 → f26368b (§11.4.26, clean fast-forward, §11.4.113-safe)** — pulled §11.4.140 v2 (BACKGROUND action), §11.4.142 (universal code-review — already actively enforced this session; added to CLAUDE.md propagated clauses), §11.4.143 (video-streaming real-user-journey — latent/N-A to this torrent-proxy project). Cascade gate **18/18 green** at the new pin.

## Open queue (Issues.md)

- **BOB-008** — RuTracker CAPTCHA — **OPERATOR-BLOCKED**. Unblock procedure documented at docs/qa/BOB-008/operator\_runbook.md (preferred: paste `bb_session` via POST `/api/v1/auth/rutracker/cookie-login`).
- Everything else closed (DB↔MD in sync, **63 items**, bin/workable-items validate OK).

## Known low-priority follow-up

- ~~pirateiro test-isolation defense in-depth~~ — **RESOLVED this session** (BOB-063, see above).

- ~~test\_credential\_env\_wiring/env\_loader\_stub-teardown-quirk~~ — **RESOLVED this session** (see above).
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## CURRENT STATE — Session 9 (2026-06-10, overnight autonomous loop)

The unit suite is now **FULLY GREEN** and **DETERMINISTIC**.

| Metric                 | Value                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Unit tests             | <b>4121 passed, 0 failed</b> —<br>pytest tests/unit/ --import-mode=importlib (5m24s)                                              |
| Determinism (§11.4.50) | top-level scope <b>3150 passed</b> identical across --<br>randomly-seed<br>7/42/100/31337/12345                                   |
| Commits this session   | 6e15a8d (crash guards +<br>async/loop hangs), 6230865<br>(test-pollution + network<br>timeouts) — both pushed to<br>all upstreams |
| Code review            | GO (zero findings/warnings)<br>after §11.4.134 iterate-until-<br>GO                                                               |

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### What landed (Session 9)

- **8 product fixes** (6e15a8d): degenerate-input crash guards for tokyotoshokan/kickass/yts/piratebay; enricher full-suite-hang fix (aiohttp.ClientTimeout ×6); kickass/bitsearch/torrentgalaxy unbounded-loop caps (MAX\_PAGES=50); mutation-scanner .venv scope fix; +7 download\_proxy tests. Closed **BOB-060**.
- **Test-suite stabilization** (6230865): eliminated all §11.4.50 order-dependent pollution — tests/conftest.py \_CORRECT\_MS\_PATH repo-root fix (the dominant bug: corrupted merge\_service.\_\_path\_\_ → broke 11 tests incl. all scheduler\_api), extended \_POLLUTING\_ROOTS, + per-test socket.socket & os.environ snapshot/restore. Network-I/O timeout hardening (search.py/routes.py/helpers.py/eztv.py). Closed **BOB-061**, **BOB-062**.

## Open queue (Issues.md)

- **BOB-008** — RuTracker CAPTCHA — **OPERATOR-BLOCKED**: needs you to complete the CAPTCHA at `/api/v1/auth/rutacker/captcha + /login`, OR paste a fresh `bb_session` cookie via `/auth/rutacker/cookie-login`. Cannot be solved autonomously.
- Everything else closed (DB↔MD in sync, 62 items, `bin/workable-items` validate OK).

## How to run the suite (both work now)

- Monolithic: `.venv/bin/python -m pytest tests/unit/ -q --import-mode=importlib` → 4121 passed.
- Per-scope (faster, parallelizable): `tests/unit/merge_service/ (801) · tests/unit/api_layer/ (170) · tests/unit/*.py` top-level (3150).
- Always pass `--import-mode=importlib` (the `merge_service.deduplicator` lazy import needs it; a bare per-file run errors).

## Known low-priority follow-up (NOT blocking — suite is green)

- A residual `test_credential_env_wiring / env_loader sys.modules-stub` ordering quirk surfaces only in narrow isolated file-combos; it does NOT manifest in the full top-level scope (5 seeds green) because the new `env/sys.modules` isolation covers it there. Tighten the `env_loader` stub teardown if it ever resurfaces.

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## Session 8 Summary

| Metric             | Value                                                                      |
|--------------------|----------------------------------------------------------------------------|
| Test files         | 47 test modules (222 <code>test_*.py</code> files total across all suites) |
| Test cases         | 4,074+ passed                                                              |
| Coverage (unit)    | 88%+ (gate: 49%)                                                           |
| Bugs fixed         | 21 (BOB-001 through BOB-025, minus BOB-008)                                |
| Submodules aligned | constitution, challenges, containers, helixqa, jackett                     |
| Ruff / Mypy        | pre-existing warnings only, 0 new                                          |

## Bug Tracker

21 bugs fixed across Session 8: - 10 B-substring parsing fixes - 3 import re missing fixes - 2 comma-size parsing fixes - 2 crash guards - 1 bt4g hang fix - 1 dedup circular import fix - 1 env\_loader flaky test fix - 1 pirateiro full test suite (44/44)

**1 remaining:** BOB-008 (RuTracker CAPTCHA) — operator-blocked, needs manual cookie paste.

## Test File Inventory

tests/unit/merge\_service/ (31 files)

- test\_cors\_config.py
- test\_dead\_tracker\_bucket.py
- test\_deadline\_tunable.py
- test\_deduplicator.py
- test\_deduplicator\_edge.py
- test\_diag\_no\_stale\_leakage.py
- test\_edge\_case\_challenges.py
- test\_enricher.py
- test\_enricher\_edge.py
- test\_hooks.py
- test\_html\_parsers.py
- test\_jackett\_autoconfig.py
- test\_nnmclub\_session\_login.py
- test\_private\_tracker\_html\_fixtures.py
- test\_public\_plugin\_harness.py
- test\_public\_plugin\_harness\_broad.py
- test\_public\_tracker\_capture.py
- test\_public\_tracker\_subprocess\_timeout.py
- test\_quality\_detection.py
- test\_scheduler.py
- test\_scheduler\_coverage.py
- test\_search\_concurrency.py
- test\_search\_coverage.py
- test\_search\_deep\_coverage.py
- test\_search\_error\_paths.py
- test\_session\_encryption.py
- test\_theme\_endpoint.py
- test\_tracker\_stats.py
- test\_ttl\_caches.py
- test\_validator.py
- test\_validator\_coverage.py

tests/unit/api\_layer/ (14 files)

- test\_auth\_coverage\_extra.py
- test\_concurrent\_writers.py
- test\_cors\_config.py
- test\_hooks\_coverage.py
- test\_hooks\_endpoints.py

test\_hooks\_remaining.py  
test\_nnmclub\_auth\_endpoints.py  
test\_routes\_coverage.py  
test\_sse\_disconnect.py  
test\_sse\_token\_auth.py  
test\_theme\_state\_coverage.py  
test\_theme\_stream.py  
test\_tracker\_stats\_sse.py

tests/unit/ (55 files, infrastructure +  
plugin tests)

test\_auth.py, test\_auth\_coverage.py, test\_auth\_models.py  
test\_config.py, test\_main.py, test\_dashboard.py  
test\_content\_type\_refinement.py  
test\_credential\_env\_wiring.py  
test\_download\_proxy\_coverage.py, test\_download\_proxy\_deep.py  
test\_download\_merged.py  
test\_env\_loader.py  
test\_freeleech.py  
test\_graceful\_shutdown.py  
test\_helpers.py  
test\_log\_filter.py  
test\_merge\_trackers.py  
test\_novaprinter.py  
test\_private\_tracker\_search.py  
test\_public\_tracker\_subprocess.py  
test\_retry\_policy.py  
test\_routes.py, test\_routes\_coverage.py  
test\_scheduler\_api.py  
test\_sorting\_weights.py  
test\_sse\_disconnect.py  
test\_streaming.py  
test\_theme\_injector.py, test\_theme\_wiring.py  
test\_webui\_theme\_injector.py  
test\_ui\_module.py, test\_ui\_sorting.py  
test\_plugin\_\*.py (39 plugin modules: eztv,  
piratebay, solidtorrents,  
limetorrents, torlock,  
gamestorrents, nyaa, kickass,  
anilibra, torrentgalaxy, yts,  
rutor, tokyotoshokan,  
snowfl, torrentdownload,  
linuxtracker, kinozal,  
nnmclub, rutracker, megapeer,  
jackett, audiobookbay,  
one337x, extratorrent,  
torrentfunk, torrentproject,  
therarbg, academictorrents,  
ali213, yourbittorrent,  
glotorrents, pctorrent,  
rockbox, bitru, btsow,  
torrentscsv, xfsb, yihua,



```

pirateiro, bt4g, iptorrents)
  test*_guards.py, test*_deep.py (crash guards + deep coverage
variants)
  test_ci_infra.py, test_ci_workflows.py
  test_community_plugins_compile.py
  test_courses_scaffold.py, test_course_scripts_lint.py
  test_docs_presence.py
  test_frontend_spec_coverage.py
  test_install_plugin_json_config.py
  test_jackett_integration.py, test_jackett_plugin_pool.py
  test_nnmclub_config_selfheal.py, test_nnmclub_plugin_login.py
  test_no_runtime_service_skips.py
  test_openapi_frozen.py
  test_page_title.py, test_footer_restored.py
  test_palette_catalog.py, test_palette_catalog_python_mirror.py
  test_quality_compose.py, test_quality_detection.py
  test_readme_landing_page.py
  test_runtime_requirements_includes_new_deps.py
  test_scan_script_non_interactive.py
  test_scanner_configs.py
  test_service_fixtures.py
  test_shadow_tokens.py
  test_socks_udp.py
  test_start_sh_copy_plugins_framework.py
  test_toolchain_config.py
  test_tracker_validator.py
  test_website_config.py
  test_branding_assets.py
  test_architecture_diagrams.py
  test_build_releases_non_interactive.py
  tests/benchmark/ (3)
  tests/chaos/ (2)
  tests/concurrency/ (3)
  tests/contract/ (5)
  tests/docs/ (3)
  tests/e2e/ (7)
  tests/integration/ (19)
  tests/memory/ (1)
  tests/observability/ (2)
  tests/performance/ (1)
  tests/property/ (2)
  tests/security/ (10)
  tests/stress/ (2)

```

## Commits (latest 10+)

```

c8254ee fix: 21 test failures, pirateiro 44/44, coverage gate 88%,
dedup circular import
ccd0dc9 fix: env_loader flaky test + coverage baseline 88.14% +
minor test fixes
3d1819c docs: finalize Session 8 – Fixed.md Rev 12, 59 items, 18

```

bugs, 41 plugins  
f924841 test: wave 8 – btsow/torrentscsv/xfsb/yihua (170 tests) +  
bt4g fix + 2 fixes  
6773203 test: wave 7 – yourbittorrent/glotorrents/pctorrent/  
rockbox/bitru (164 tests) + bitru fix  
f191401 test: wave 6 – torrentfunk/torrentproject/therarbg/  
academictorrents/ali213 (178 tests) + 2 fixes  
1446cc5 test: wave 5 – audiobookbay/one337x/extratorrent (155  
tests) + 3 fixes  
fd543b0 docs: update CONTINUATION.md for Session 8 (4 waves, 23  
plugins)  
aae4069 test: wave 4 – kinozal/nmclub/rutracker/megapeer/jackett  
+ megapeer B-substring fix  
e3b7acb test: wave 3 – rutor/tokyotoshokan/snowfl/torrentdownload/  
linuxtracker + nyaa/kickass fixes

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## Coverage Snapshot

| Module                | Coverage    |
|-----------------------|-------------|
| api/init.py           | 98%         |
| api/auth.py           | 91%         |
| api/routes.py         | 95%         |
| api/streaming.py      | 99%         |
| main.py               | 94%         |
| deduplicator.py       | 94%         |
| enricher.py           | 100%        |
| hooks.py              | 95%         |
| jackett_autoconfig.py | 99%         |
| scheduler.py          | 93%         |
| search.py             | ~90%        |
| validator.py          | 92%         |
| theme_injector.py     | 99%         |
| env_loader.py         | 100%        |
| download_proxy.py     | ~95%        |
| <b>TOTAL (unit)</b>   | <b>88%+</b> |

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## Known Issues

1. **BOB-008**: RuTracker CAPTCHA — operator-blocked (needs manual cookie paste)
2. Go backend is a skeleton (documented in AGENTS.md)

3. Containers may be down on session start — `bash start.sh` first
  4. macOS + podman `network_mode: host` does NOT forward ports  
— `ensure-macos-tunnel.sh` handles this
- 

## Quick-Start

```
# One-time venv bootstrap (gitignored .venv/ — see docs/  
    TESTING.md "Bootstrap")  
python3 -m venv .venv  
.venv/bin/pip install --upgrade pip  
.venv/bin/pip install -r tests/requirements.txt  
  
# Pre-build gate  
bash scripts/pre_build_verification.sh  
  
# Unit tests  
.venv/bin/python -m pytest tests/unit/ -q --import-mode=importlib  
  
# Coverage  
.venv/bin/python -m pytest tests/unit/ --cov=download-proxy/src --  
    cov=plugins --cov-report=term --import-mode=importlib  
  
# Lint + typecheck  
.venv/bin/ruff check . && .venv/bin/mypy download-proxy/src/  
  
# Frontend  
cd frontend && npx vitest run  
  
# Containers  
bash start.sh && bash stop.sh
```

---

## Architecture

| Port | Service              | Tech                          |
|------|----------------------|-------------------------------|
| 7185 | qBittorrent WebUI    | LinuxServer (admin/admin)     |
| 7186 | Download Proxy       | Python HTTP                   |
| 7187 | Merge Search Service | FastAPI + Angular SPA         |
| 7188 | WebUI Bridge         | Python (host process)         |
| 7189 | boba-jackett         | Go/Gin                        |
| 9117 | Jackett              | LinuxServer (auto-configured) |